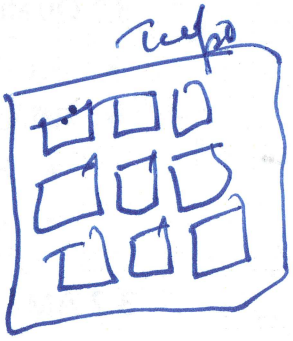
Planet Smoke plume + Tempo data
emissions

1.1 Acquire coincident data

- **PlanetScope**: surface reflectance (SR), RGB/NIR bands
- **TEMPO**: Level-2 tropospheric NO_2 vertical column density (VCD), averaging kernel (AK), air mass factor (AMF), QA flags
- Optional:
  - GOES-16 for cloud/smoke context
  - Meteorology (winds, PBL height) from NOAA or reanalysis



1.2 Temporal collocation

- Select TEMPO overpass **closest in time** (hourly).
- If plume evolves rapidly, use **two adjacent TEMPO hours** to assess temporal sensitivity.

1.3 Spatial alignment

- Reproject PlanetScope to TEMPO grid (or vice versa).
- Keep both:
  - Native high-res plume mask (Planet)
  - Coarser TEMPO pixels for NO_2

Tempo is ~ 2-4km

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2) Smoke plume detection from Planet

2.1 Derive plume mask

Use spectral tests (example):

- Blue/NIR ratio or haze index
- Texture + brightness thresholds
- Optional ML segmentation

Output:

- Binary mask: $M(x, y) \in \{0, 1\}$

2.2 Sub-pixel plume fraction (critical)

Aggregate to TEMPO grid:

$$f_p = \frac{\sum M_{\text{Planet}}}{N_{\text{Planet pixels within TEMPO pixel}}}$$

This yields **plume fraction per TEMPO pixel.**

3) Prepare TEMPO NO_2

3.1 Quality filtering

- Use QA flags (cloud fraction, retrieval quality)
- Remove:
  - High cloud fraction (> ~0.2–0.3)
  - Low QA pixels

3.2 AMF / vertical sensitivity correction

Scd ≈ VCD × AMF

TEMPO NO_2 is:

provided $\Leftarrow$ $$VCD = \frac{SCD}{AMF}$$ ✓ provided

For smoke plumes, vertical profile differs → adjust AMF:

$$AMF_{\text{adj}} = \sum w(z)\, AK(z)$$

Where:

- $AK(z)$ = averaging kernel → you can get from hdf files
- $w(z)$ = assumed NO_2 vertical profile (use plume injection height from fire data or literature)

→ 0-5km – 2km (est)

Then:

$$VCD_{\text{adj}} = \frac{SCD}{AMF_{\text{adj}}}$$

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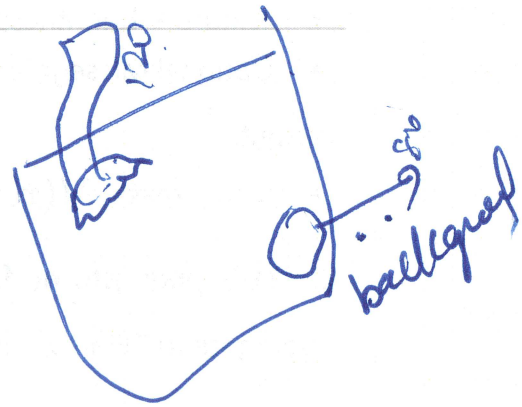
4) Isolate plume NO_2 enhancement



4.1 Background estimation

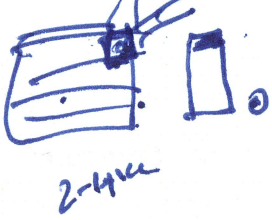
Define upwind/background region:

- Outside plume mask
- Similar surface/urban conditions

Compute:

$$VCD_{\text{bg}} = \text{median}(VCD_{\text{adj, background}})$$

✓



pixel cm

4.2 Excess NO_2

For each TEMPO pixel:

$$\Delta VCD = VCD_{\text{adj}} - VCD_{\text{bg}}$$

Apply plume fraction:

$$\Delta VCD_{\text{plume}} = f_p \cdot \Delta VCD$$

plume fraction

$f_p = \frac{\text{plume area within pixel}}{\text{total pixel area}}$

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5) Convert column to mass

For each pixel:

$$\text{Mass}_{NO_2} = \Delta VCD_{\text{plume}} \cdot A \cdot M_{NO_2}$$

Where:

- A = pixel area (m^2)
- M_{NO_2} = molecular mass conversion
- Convert molecules/cm^2 → mol/m^2 → kg

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6) Emission / flux estimation (key step)

Method A: Mass balance (box method)

Total plume mass:

$$M_{\text{total}} = \sum \text{Mass}_{NO_2}$$

Estimate emission rate:

$$E = \frac{M_{\text{total}}}{\tau}$$

Where:

- τ = effective NO_2 lifetime (~1–6 hrs depending on chemistry)

Method B: Flux (cross-section) method (preferred)

1. Define transect perpendicular to wind